Math-Augmented Praxeology

A minimal mathematical annex to the praxeological analyser

Math core rules, survey of methods, worked examples

Not praxeology: it sizes what the core rules have signed

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Introduction

A separate, minimal mathematical annex to the praxeological analyser. **It is not praxeology.** It lives in its own folder so that nothing in `handbook/` or `praxeo/` depends on it. Nothing in the main analysis may cite it as a reason for a sign.

Start with `MATH_CORE_RULES.md`. It sets out the seven math core rules (MR-1 to MR-7) that every annex function and handler is derived from, as the handbook's patterns are derived from its core rules. It also gives the admission test: a method belongs here if and only if it can be derived from MR-1 to MR-7 with sourced input ranges. `SURVEY.md` applies that test to the main mathematical methods economists use and sorts each one: adopted, adopted as a bound only, usable as an input only, or excluded, with the reason.

What it is for

The praxeological method gives directions and conditions: rents for sitting tenants fall, quantity supplied falls if owners have other uses for the buildings. It does not give sizes. The annex adds sizes and bounds to consequences the method has already signed, using as little mathematics and as few assumptions as possible.

Rules

1. **It never sets or changes a sign.** Every result is checked against the praxeological sign (`check_sign`) and reported as one of:
    - *agrees*.
    - *indeterminate*: the range straddles zero, or the sign was `ambiguous`.
    - *contradicts*: an input range is wrong, or a praxeological premise failed. Report it; never resolve it here. The annex pass names the premise each kind of contradiction points to, with its probability in the run (`bridge.DIAGNOSES`). For example, a negative price gap under a ceiling means `binding` fails.
2. **Ranges, never points.** Every input is an interval the analyst must source, and every output is an interval. Arithmetic is interval arithmetic. Bounds come from the corners only where the function is monotone in each input, and each function says why it is.
3. **The fewest assumptions that do the job, on a ladder.** Each result states its rung and lists every assumption it took.

Rung	What it assumes	Examples here
M0	Nothing beyond its inputs: identities and arithmetic.	Equation of exchange and the Fisher relation. Inflation tax, debt dynamics, sectoral balances. Present value and capitalization, price paths, stock-flow arithmetic. Little's law, expected penalties, quota rents.
M1	Local linearity only: for small changes, %ΔQ = elasticity × %ΔP, with elasticities given as ranges. No fitted curve, no functional form.	Supply and demand intersection. Ceiling shortage and floor surplus. Tax and subsidy incidence, exchanges forgone, the tariff. Spending change, cost pass-through (as a bound), tax revenue with base response.
M2 and above	Functional forms, estimated systems, general equilibrium.	**Excluded.** Too many assumptions for what they add here.
Excluded at any rung	Money measures of welfare: consumer and producer surplus, deadweight loss in money, welfare cost-benefit.	They add up ordinal value across persons. The annex counts the exchanges forgone instead.

4. **M1 says when it is strained.** Above a 15% price gap, a constant elasticity is a strong assumption, and the result carries a warning to treat it as indicative only.
5. **No hidden splits.** Where the inputs leave a quantity undetermined, the result is the full range it could take, with a note. Tax and subsidy incidence when neither side can respond is one case: buyers bear

anything from none to all of the tax. A conventional split is never assumed.

6. **Signs inside M1 come from praxeology.** Downward-sloping demand and non-negative supply response are the laws of demand and supply (LW-3, LW-4). A demand elasticity above zero is rejected, not used.

How the annex relates to the core rules

1. The core rules constrain the mathematics, never the reverse.

- Directions come from the core rules and the laws. Downward-sloping demand and a non-negative supply response are LW-3 and LW-4; a demand elasticity above zero is rejected, not used. The annex sizes those directions and derives none.
- Methods are excluded exactly where they conflict with the core rules:
  - Keynesian multipliers contradict CR-8: means taken by a tax come from other uses.
  - Surplus and deadweight loss in money contradict the ordinal value behind CR-2 and CR-3.
  - Monopoly and game models replace CR-4's ordinal margin with an assumed model of how people decide.
- A range with the opposite sign to the praxeological consequence is reported as *contradicts*. It means an input range is wrong or a praxeological premise failed. The annex never resolves it by changing the sign. Each handler records which premise its contradiction points to. The report names that premise and its probability in the run, for example "the ceiling does not bind: `binding(c1)` (p = 0.75)", so the reviewer knows where to look.
- So within its scope the annex quantifies and does not contradict. It can still change what a report says, through the premise step (point 6), and only through a decider and with a label. It also differs from Mises on whether quantitative relations in human action are stable: first-order results treat a sourced elasticity range as stable enough to size an effect. That is why an M1 result never settles a sign, and why large gaps are flagged.

2. Where each part attaches.

Core rule	What the core rule decides	What the annex can add
CR-3, CR-8 (cost; conservation of means)	Who bears a loss; that means taken come from other uses	How much: inflation tax, debt dynamics, sectoral balances
CR-4 (response at the margin)	Direction of every switch; the means exception is `ambiguous` without a premise	`spending_change` (whether spending on the good rises or falls); `expected_penalty` (evidence for an enforcement premise)
CR-5 (exchange)	Exchanges that no longer happen (CR-5a)	`quantity_under_wedge`: how many, counted, never valued
CR-6 (price formation, displaced pressure)	Shortage, surplus, and pressure moved to other margins	Shortage, surplus, equilibrium shift, tax and subsidy incidence, wait times (Little's law)
CR-7 (equalization across uses; CR-7c)	Resources move; asset values change when the policy is expected	Present value, capitalized loss, quota and licence values
CR-9 (propagation)	The chain to linked markets	Cost pass-through (a bound); imports as use minus domestic output
CR-10 (time)	Timing per window	Price-gap paths (when a cap starts to bind), stock-flow (how fast a durable stock shrinks), doubling times
CR-12 (composition)	Opposing chains combine to `ambiguous` unless arithmetic or a premise settles them	See point 3

CR-1, CR-2 and CR-11 (what a policy can do to an actor, choice, knowledge) have no counterpart here, and should not have one.

3. CR-12 is the one point of contact, and it is guarded. CR-12e (0.8) already allows opposing chains to be combined by arithmetic where the case gives the figures, and by a premise otherwise. The annex respects that line:

- **M0 on the case's own figures may settle.** `bridge.settle_by_arithmetic` lets an M0 result settle a CR-12 comparison only when every input it used comes from the case record (`from_case: true`). That is the arithmetic CR-12e already allows, done in code.
- **M1, or any result on illustrative or outside ranges, can only be evidence.** `bridge.premise_evidence` proposes a probability for the dominance premise, with its source:
    - one range wholly larger than the other gives 0.9 or 0.1;
    - overlapping ranges give no change.

    It never writes the probability. A decider (LLM, human or Jev) must accept it, and the source is logged. `settle_by_arithmetic` raises `GuardError` if it is given such a result.

Without this guard, an elasticity guess could decide a sign through the back door.

4. Two labels, never merged. A consequence keeps its tier (A, B, J, C), which says how its sign was derived. An annex result carries a rung (M0, M1), which says what its size assumes. A Tier A sign with an M1 size is Tier A for its sign; the size is only as good as its ranges.

5. One-way dependency, checked. `bridge.annex_pass` reads a frozen `prediction.json` from `praxeo`. For each reported consequence whose variable has a handler, it attaches a range, its rung, the sign check and the source of every input. It verifies that the prediction is unchanged and raises `GuardError` if not. Nothing in `praxeo/` or `handbook/` imports the annex.

```
python3 -m praxeo cases/sf_1994 --out runs/sf
python3 math-augmented-praxeo/annex_run.py runs/sf/prediction.json math-augmented-
praxeo/examples/sf_1994/annex_inputs.json runs/sf \
    --premise-step --decider fixture --answers math-augmented-
praxeo/examples/sf_1994/premise_answers.json
```

`examples/sf_1994/annex.md` is the result for the committed sample run. It attaches sizes to three consequences, and all three agree with their signs:

- the covered stock's quantity supplied;
- sitting tenants' rent against the market path;
- the shut-out seekers' price for uncovered units.

All ranges except the horizon are illustrative.

6. The premise step. `premise_step.run` takes each dominance premise that the inputs can size (`_dominance` in the inputs file names the counts behind each chain). When the annex proposes a different probability, it asks a `praxeo` Decider a typed noul question: fixture, LLM, human or later Jev. The question states the premise, its current probability, the annex evidence with its rung, and the proposal. The decider's answer becomes the premise's probability, whatever the proposal was. The step then re-derives the prediction with the `praxeo` engine, combiner and verifier, and writes it as `prediction_annex_informed.json` beside the frozen original, which stays the scored one. Every consequence whose sign or reported status differs is listed and labelled "annex-informed", with the evidence.

In `examples/sf_1994/premise_step.md`:

- **keepers_dominate:** the annex proposed raising it from 0.60 to 0.90, because the deterred exits exceed the displaced exits over the whole range. The fixture decider gave 0.80. No consequence changed, because the premise was already above the cut-off.
- **lockin_dominates:** its ranges overlap, so no change was proposed and the decider was not asked.

The tests also cover a premise at 0.4. There the decider's acceptance turns the class total from `ambiguous` to `decrease`, labelled as annex-informed; a decider that declines leaves it `ambiguous`.

The equation of exchange

M V = P Y is used as an **identity** (rung M0). In growth factors it holds exactly: (1 + p) = (1 + m)(1 + v) / (1 + y). The identity is close to a priori: it restates that spending equals spending. The *quantity theory* goes further

and assumes that velocity and real output do not respond to money. That claim is empirical, so here it enters only as ranges the analyst supplies for v and y.

Mises rejected the equation as an explanation (*Theory of Money and Credit*, Part II ch. 8; *Human Action* ch. XVII §4). P is an average, and new money reaches some people first. So individual prices rise at different times and by different amounts, and relative prices and wealth shift along the way (Cantillon effects). The annex therefore bounds the average only. Who gains and who loses stays a praxeological chain.

Modules

Module	Rung	Contents
`mathpraxeo/interval.py`	—	Interval arithmetic and corner evaluation for monotone functions.
`mathpraxeo/result.py`	—	`Result` (range, rung, assumptions, note) and `check_sign`.
`mathpraxeo/exchange.py`	M0	Price-level change from m, v and y, and the implied v from observed m, p and y.
`mathpraxeo/identities.py`	M0	Little's law (wait time), expected penalty, inflation tax, Fisher relation, debt dynamics, stock-flow, sectoral balances, doubling time.
`mathpraxeo/capital.py`	M0	Gap path between a capped and a market price, the gap as a share of the market price, the capitalized loss, and the present value of a yearly amount (for license, quota and tax values).
`mathpraxeo/statics.py`	M1 (quota rent: M0)	Supply and demand intersection; ceiling shortage, floor surplus, quantity supplied under a held-down price; tax and subsidy incidence; exchanges forgone; tariff; quota rent; spending change; cost pass-through bound; tax revenue with base response.
`mathpraxeo/bounds.py`	M1, bounds only	Black-market premium under a ceiling, Hotelling timing, rent-seeking ceiling, covered interest parity. Each returns limits, never a point.
`mathpraxeo/rules.py`	—	The math core rules as data, their praxeological counterparts, and the `@derives(...)` decorator that records each function's derivation line.
`mathpraxeo/report.py`	—	Renders an annex table with the sign check for each consequence.
`mathpraxeo/bridge.py`	—	The annex pass over a `praxeo` prediction, with handlers keyed by instrument, variable and actor role, and its guards: `annex_pass`, `dominance_evidence`, `premise_evidence`, `settle_by_arithmetic`, `GuardError`.
`mathpraxeo/premise_step.py`	—	Puts annex evidence for dominance premises to a decider; writes the log and the annex-informed prediction.
`annex_run.py`	—	Command line for the annex pass and the premise step.

Examples

The elasticities, shares, rates and durations in both examples are illustrative ranges, not estimates.

- **`examples/we001_annex.py`** (WE-001, Riverton rent freeze) sizes three consequences: the year-3 shortage, the year-10 fall in quantity supplied, and the fall in building values at announcement. All three ranges agree with the praxeological signs. Two are flagged as strained, because the gaps grow large.
- **`examples/we013_annex.py`** (WE-013, Halvern steel tariff of 25%) sizes six consequences. Five agree with their signs:
    - the domestic steel price;
    - domestic output;
    - imports;
    - steel bought;
    - revenue short of the static projection.

  The sixth, construction prices, is *indeterminate*, because a pass-through bound includes zero. That is correct: the sign comes from the praxeological chain, and the annex only bounds its size.

Run

```
python3 -m pytest math-augmented-praxeo/tests
python3 math-augmented-praxeo/examples/we001_annex.py
```

Status and next steps

- This is a folder on the working branch, not a separate branch. A separate branch can be cut from it if wanted.
- The annex pass has 27 handlers, keyed by instrument, variable and actor role. They cover:
    - price ceilings: quantity supplied, shortage, insiders' price, capitalized asset value, the substitute's price;
    - floors: surplus, employment and quantity demanded;
    - per-unit taxes: both prices, quantity traded, revenue shortfall;
    - tariffs: domestic price, domestic output, imports, use, revenue shortfall;
    - subsidies: both prices;
    - quotas, licences and allocated rights: the right's value and its lease price;
    - for any instrument: wait times (Little's law) and cost pass-through to a downstream buyer;
    - the bounds: the illegal premium under a ceiling (`side_payments`), the timing of extraction under a resource tax or quota (`resource_stock`), and the ceiling on lobbying spending (`lobbying_effort`). A bound that includes zero gives *indeterminate*, never *agrees*.

  Actor roles not named in the record (buyers, sellers, importers, holders, downstream) come from `_actors` in the inputs file. The thin slice only produces price-ceiling consequences, so the other handlers are exercised by tests on synthetic predictions.
- The premise step covers the two dominance premises the rule base has (`keepers_dominate`, `lockin_dominates`). New dominance premises need an entry in `DOMINANCE_PREFIXES` and a `_dominance` spec.
- The LLM decider has not run here (no API key).
- Covered interest parity is a function only (`bounds.covered_interest_parity`): the ontology has no forward-rate variable for a handler to attach to.

Math core rules

The annex's counterpart to the handbook's core rules (`handbook/04a_core_rules.md`). The handbook derives its 106 patterns from twelve core rules. The annex derives every one of its functions and handlers from the seven rules below. Each function and handler carries a derivation line naming them: `@derives(...)` in the code, and `HANDLER_RULES` in `mathpraxeo/bridge.py`. A test checks that none is missing.

These rules are not praxeology. Each says what it takes as given and which praxeological core rule it is the quantitative counterpart of. None of them can produce a sign. Signs come only from the core rules. The math rules size what the core rules have already signed, or rank two magnitudes under the guard in MR-7.

The admission test

> A method belongs in the annex if and only if it can be derived from MR-1 to MR-7, with every input a sourced range.

`SURVEY.md` judged each method one by one. The test turns those judgments into a rule. Every excluded method fails it at an identifiable point (see "Verdicts restated" below). A method proposed later is admitted by writing its derivation line, or excluded by naming the step no rule covers. This is the same discipline the handbook applies to a new pattern.

Rungs

- **MR-1, MR-2, MR-3 are exact given their inputs.** They are identities, definitions and arithmetic, so their results are rung **M0**.
- **MR-4 and MR-5 assume a stable response over the change considered.** Their results are rung **M1**, and this is where the annex parts from Mises: there are no constant quantitative relations in human action.
- **MR-6 sets limits.** Its direction is praxeological: switching narrows a gap (CR-4, CR-7). How far the gap can stay open is a size, and it depends on switching costs given as ranges. Results that rest on it are **bounds**.
- **MR-7 governs how magnitudes may combine.** It is the annex's form of CR-12.

MR-1 Ranges

Statement. Every quantity is a set of possible values: a closed interval given by its lowest and highest value. Operations carry sets to sets, and a result is the full set its inputs allow. A function monotone in each input separately reaches its extremes at the corners of the input box, so corner evaluation is exact for it. Where an input range includes a point at which the result is undetermined, the result is the full range of values it could take there. No conventional value is chosen ("no hidden splits").

Exact or assumed. Exact: it is interval arithmetic. The only assumption is that each input range contains the true value, which is the input's source's claim, not the annex's.

Praxeological counterpart. The method's rule that no premise stands without an estimate: an assumption gets a probability, and a quantity gets a range.

Limits. Corner evaluation is used only where monotonicity in each input has been checked. Otherwise a grid that includes the corners is used (`statics._extremes`), and the function says so.

Used by. Everything (`mathpraxeo/interval.py`). It is implied and not named in derivation lines.

MR-2 Identities and conservation

Statement. Over a closed set of accounts, flows balance, and quantities defined as sums, differences or products of others equal them. Spending equals receipts (M V = P Y). Saving net of investment sums to zero

across the private, government and foreign sectors. A stock changes by its inflows minus its outflows. Imports are domestic use minus domestic output. A queue's average length is its arrival rate times the average wait (Little's law). A right's yearly rent is the price gap times the quantity it covers. An expected cost is the probability times the amount.

Exact or assumed. Exact, by definition or by counting. The identity says nothing about which account moves when another changes: that is a causal question for the core rules.

Praxeological counterpart. CR-8 (conservation of means): means taken by a policy come from other uses. MR-2 does the counting; CR-8 says whose means.

Limits. An identity used as a theory is outside the annex. For example, the quantity *theory* of money, which says velocity and output do not respond to money, enters only as ranges for v and y. Aggregates such as the price level are averages; who gains and who loses stays a praxeological chain (Cantillon effects).

Used by.

- `exchange`: the price-level change and the implied velocity change.
- `identities`: wait time, expected penalty, inflation tax, sectoral balance; debt and stock paths (with MR-3).
- `statics`: quota rent; imports in the tariff; the accounting in spending and revenue changes.
- `bridge.dominance_evidence`: the counts behind each chain.

MR-3 Compounding and discounting

Statement. A value growing at rate g for t periods is multiplied by (1 + g)^t. An amount due in t periods is worth 1 / (1 + r)^t of itself today, at discount rate r. Present values add over periods. The Fisher relation (1 + i) = (1 + r)(1 + π) defines the real rate.

Exact or assumed. Exact given the growth rates, the discount rate and the duration. All three are inputs, given as ranges.

Praxeological counterpart.

- CR-10 (time): when an effect appears and how it grows.
- CR-7c: a lasting change in income is priced into an asset when it becomes expected. MR-3 says by how much, given the buyers' discount rate and expected duration.
- Time preference (LW-7 in the foundations) is why the discount rate is positive, but the annex takes its size as an input.

Limits. The discount rate and expected duration are the buyers' (CR-7c). They are estimated as ranges, never derived.

Used by.

- `capital`: gap paths, gap shares, capitalized loss, present value.
- `identities`: real rate, doubling time; the debt and stock paths (with MR-2).
- Handlers for insiders' price, asset values and rights' values.
- Inside the bounds: Hotelling and the rent-seeking ceiling.

MR-4 First-order response

Statement. For a small change in price, the proportional change in quantity demanded or supplied is the elasticity times the proportional change in price. The elasticity's **sign** comes from the laws: demand falls as price rises (LW-3), and supply does not fall (LW-4). A demand elasticity above zero is rejected. The elasticity's **size** is a sourced range for the market and horizon in question. Products of a price and a quantity (spending, revenue) change, to first order, by the sum of their proportional changes.

Exact or assumed. Assumed: the response is stable enough over the change for one elasticity range to describe it. This is the annex's first departure from praxeology. Mises held there are no constant relations in human action, so the range is history, not law. Results beyond a 15% price gap are flagged as strained.

Praxeological counterpart. CR-4 (response at the margin). CR-4 gives the direction and says that some actors are at the margin (the `substitutability`, `alternative_uses` and `mobility` premises). MR-4 sizes how many.

Limits. It is local only. It never decides CR-4's means exception (the direction of spending when the price of a heavily bought good changes); it can only size each side for the premise. It never values the exchanges forgone, only counts them.

Used by. In `statics`: ceiling shortage, quantity supplied under a held-down price, floor surplus, quantity under a wedge, spending change, revenue change, and the tariff's output and use. Handlers built on these, and the black-market upper limit.

MR-5 Clearing as a tendency

Statement. When demand shifts by a share a and supply by a share b at the old price, the first-order price change that would remove the excess is (a − b) / (εs − εd). The matching quantity change is (εs·a − εd·b) / (εs − εd). A wedge (a tax or subsidy) between buyers' and sellers' prices splits between them in proportion to the opposite side's response. Buyers bear εs / (εs + |εd|) of it.

Exact or assumed. Assumed: MR-4's stable responses, and that the price moves toward clearing. It measures the size of a tendency, not a state the market reaches. Equilibrium is an imaginary construct in Mises (the evenly rotating economy): the annex uses where the curves would meet only to size the pressure.

Praxeological counterpart. CR-6 (price formation and displaced pressure): a price held off its clearing level displaces pressure to other margins. CR-6 also says which side bears a wedge, according to which side can move more. MR-5 says how much.

Limits. It never claims the market reaches the new point. If both responses can be zero, MR-1's no-hidden-splits rule applies: the share is anything from none to all.

Used by. `statics`: equilibrium shift, tax incidence, subsidy incidence. Handlers for the substitute's price and the tax and subsidy prices.

MR-6 No arbitrage

Statement. Where actors can switch between two options (two markets, two times, two currencies, legal and illegal trade, owning and leasing a right), the difference in what the options yield cannot stay above the cost of switching. If it did, someone would gain by switching until it no longer did. So:

- prices and yields are bounded by each other plus switching costs;
- a pass-through lies between none and all;
- an illegal premium lies between what sellers must be paid for the risk and what buyers will pay;
- a resource is extracted sooner when its net price grows more slowly than the rate of interest;
- spending to win a privilege does not exceed its value.

Exact or assumed.

- The direction is praxeological: actors switch toward the option that yields more (CR-4), which equalizes yields across uses (CR-7).
- The bound's position depends on switching costs, penalties and rates, which are inputs given as ranges.
- That arbitrage has finished is not assumed: the bound says where the gap must end up, not when.

Praxeological counterpart. CR-7 (equalization across uses) and CR-4. MR-6 is the quantitative shadow of CR-7.

Limits. It gives bounds, not points. A bound that includes zero cannot confirm a sign, so the sign check reports *indeterminate*.

Used by.

- `bounds`: black-market premium, Hotelling gap, rent-seeking ceiling, covered interest parity.
- `statics`: cost pass-through; the tariff's pass-through range.

- Handlers for tariff prices, lease prices of rights, and the bounds.

MR-7 Composition of magnitudes

Statement. Two opposing chains are ranked by size only when their ranges do not overlap. If they overlap, the comparison is undetermined. A ranking may *settle* a CR-12 comparison only when both sizes are rung M0 and every input comes from the case record. That is the arithmetic CR-12e (0.8) already allows. In every other case, the ranking is *evidence* for a dominance premise: the annex proposes a probability, and a decider decides. A sign is never set by MR-7.

Exact or assumed. Exact as a comparison of sets. The guard is a rule of method, not a mathematical fact: it keeps M1 assumptions from deciding a sign.

Praxeological counterpart. CR-12 (composition): opposing chains combine to `ambiguous` unless arithmetic or a premise settles them.

Used by. `bridge.premise_evidence`, `bridge.dominance_evidence`, `bridge.settle_by_arithmetic` (raises `GuardError` when the guard is breached), and `premise_step`.

Derivation lines

Functions.

Module	Function	Math rules
exchange	price_level_change, implied_velocity_change	MR-2
identities	wait_time, expected_penalty, inflation_tax, sectoral_balance	MR-2
identities	debt_path, stock_path	MR-2, MR-3
identities	real_rate, doubling_time	MR-3
capital	cap_gap_path, gap_share, capitalized_loss, present_value	MR-3
statics	ceiling_shortage, ceiling_quantity_supplied, floor_surplus	MR-4
statics	equilibrium_shift, tax_incidence, subsidy_incidence	MR-5
statics	quantity_under_wedge	MR-4, MR-5
statics	quota_rent	MR-2
statics	spending_change, revenue_change	MR-2, MR-4
statics	tariff	MR-2, MR-4, MR-6
statics	cost_pass_through	MR-6
bounds	black_market_premium	MR-4, MR-6
bounds	hotelling_gap, rent_seeking_ceiling	MR-3, MR-6
bounds	covered_interest_parity	MR-6
bridge	premise_evidence, settle_by_arithmetic	MR-7
bridge	dominance_evidence	MR-2, MR-7

Handlers. The derivation lines of the 27 handlers are in `HANDLER_RULES` (`mathpraxeo/bridge.py`). They use MR-2 to MR-6 only, and each of the five at least once. MR-7 is reserved for the CR-12 machinery. The annex pass's report shows each handler's rules beside its result.

Verdicts restated

The verdicts of `SURVEY.md`, restated against the rules. "Admitted" gives the derivation line. "Fails at" names the step that no rule covers.

Method	Verdict	Derivation, or where it fails
Equation of exchange; sectoral balances; balance of payments	Admitted	MR-2

Method	Verdict	Derivation, or where it fails
Quantity theory of money (as a theory)	Fails at: a behavioural claim (v and y do not respond to M)	Usable only as input ranges for v and y
Fisher relation; present value; capitalization; compounding	Admitted	MR-3
Debt dynamics; stock-flow	Admitted	MR-2, MR-3
Little's law; expected penalty; inflation tax; quota rent	Admitted	MR-2
Supply and demand intersection; incidence	Admitted	MR-5 (with MR-4)
Shortage, surplus, quantity under a wedge, spending and revenue change	Admitted	MR-4 (with MR-2)
Tariff	Admitted	MR-2, MR-4, MR-6
Cost pass-through; black-market premium; Hotelling; rent-seeking ceiling; covered interest parity	Admitted as bounds	MR-6 (with MR-3, MR-4)
Consumer and producer surplus; deadweight loss in money; welfare cost-benefit; compensating variation	Fails at: summing value across persons in money	No rule adds ordinal value. MR-4 counts the exchanges forgone instead
Utility maximization with calculus	Fails at: a cardinal or differentiable utility	CR-4 gives the directions without it
Production functions (Cobb-Douglas, CES)	Fails at: a functional form	Cost shares are admitted as inputs (MR-6 pass-through)
Perfect competition as a benchmark	Fails at: distance from a state that never obtains	MR-5 sizes tendencies from the actual position
Monopoly markup (Lerner)	Fails at: a known demand curve and an assumed goal (profit maximization at a point)	—
Oligopoly models; Nash equilibria needing common knowledge	Fails at: an assumed model of how rivals think	Dominant-strategy arguments are CR-4 already
Search and matching; cobweb models	Fails at: matching functions; an assumed way of forming expectations	Little's law (MR-2) gives waiting arithmetic
Keynesian multipliers	Fails at: a fixed spending propensity treated as a law, and spending as the source of income	Contradicts CR-8; MR-2's identities do not imply it
IS-LM, AD-AS, Phillips curve, Taylor rule	Fails at: aggregate behavioural relations	—
DSGE and CGE models	Fails at: functional forms and calibrated parameters	—
Input-output (Leontief) as a model	Fails at: fixed coefficients	Its cost shares are admitted as inputs
Growth accounting	Fails at: a production function residual	—
Ricardian equivalence	Fails at: a behavioural claim (people offset deficits by saving)	The debt arithmetic (MR-2, MR-3) is admitted
Purchasing power parity	Fails at: a long-run tendency stated as a law	Usable as an input range
Econometrics, forecasts, index numbers	Not methods of derivation	Sources for input ranges

Survey of mathematical methods

This file goes through the main mathematical methods economists use. The verdicts are restated against the math core rules, as derivation lines or the step that fails, in `MATH_CORE_RULES.md` ("Verdicts restated"). For each it says whether the method meets the annex's rules (README) and whether it is relevant to the handbook's patterns. A method qualifies when it meets three conditions:

- It needs no assumption beyond identities, arithmetic or local linearity with sourced elasticity ranges.
- It never has to decide a sign.
- It adds a size, a bound or a timing the praxeological chain lacks.

Verdicts

- **Adopt:** in the annex.
- **Bound:** adopted only as an upper or lower bound, never as a point estimate.
- **Input:** not a method of derivation; a legitimate source for the ranges the annex takes as inputs.
- **Exclude:** fails the rules, with the reason given.

Rungs

- **M0:** identities and arithmetic.
- **M1:** first-order, local, with elasticity ranges.

1. Accounting identities and arithmetic (M0)

Method	Verdict	Assumes	Relevance	Module
Equation of exchange, MV = PY	Adopt, as an identity	Nothing; v and y enter as ranges	Monetary policies, budget deficits financed by money, the price level (TX-8, OC-8)	`exchange.price_level_change`
Quantity theory of money (v and y independent of M)	Input	That velocity and output do not respond to money: empirical	Same as above; enters only as narrow ranges for v and y	(ranges for `exchange`)
Fisher relation: (1+i) = (1+r) (1+π)	Adopt, as a definition	Nothing (ex post); expected inflation is an input	Real rates under interest caps and inflation; debt policies	`identities.real_rate`
Inflation tax on money balances	Adopt	Nothing beyond the inputs	Who loses when prices rise: holders of money. Pairs with Cantillon effects	`identities.inflation_tax`
Government budget constraint and debt dynamics	Adopt	Nothing; r, g and the primary balance are ranges	OC-8 (crowding out), OC-9 (debt-funded transfers), TX-9	`identities.debt_path`
Sectoral balances: (S−I) + (T−G) + (M−X) = 0	Adopt, as an identity	Nothing	Deficits and the currency link (TX-8, OC-8). The identity says nothing about which balance adjusts	`identities.sectoral_balance`
Balance of payments (current + capital account = 0)	Adopt, as an identity (same logic as sectoral balances)	Nothing	TX-8 (imports provide the means to buy exports or assets)	via `sectoral_balance`
Present value and capitalization	Adopt	A discount rate and an expected duration, as ranges	CR-7c, SB-4 (asset values), value of licenses and quotas (EN-), lasting taxes	`capital.capitalized_loss`, `capital.present_value`

Method	Verdict	Assumes	Relevance	Module
Price-path arithmetic (caps, indexation, compounding)	Adopt	Growth rates as ranges	PC-1, TS-3, TS-10 (when a cap starts to bind; size of the gap)	`capital.cap_gap_path`, `identities.doubling_time`
Stock-flow accounting: K' = K(1−δ) + additions	Adopt	Depreciation and additions as ranges	TS-1, PC-5 (how fast a durable stock shrinks when building stops)	`identities.stock_path`
Little's law, L = λW (queueing)	Adopt	A stable queue	PC-2 (rationing by waiting), public supply at a set price (PC-19)	`identities.wait_time`
Expected value of a penalty	Adopt, to inform premises	Detection probability and penalty as ranges	Enforcement premises (A-2-type), RS-3, RS-20. Informs a premise; never decides it	`identities.expected_penalty`
Index numbers (CPI, deflators)	Input	The choice of basket	Converting nominal to real figures. Mises's critique applies (an average of incommensurable changes), so indices are inputs, never conclusions	—
Comparative advantage arithmetic (Ricardo)	Already praxeological	—	Mises's law of association (HA ch. VIII). The arithmetic illustrates a priori reasoning and adds nothing to it	—
Lerner symmetry (an import tax equals an export tax)	Already praxeological	Trade balanced over time	TX-8 already derives it through the currency link	—

2. Partial equilibrium: supply and demand, first order (M1)

Method	Verdict	Assumes	Relevance	Module
Supply and demand intersection; comparative statics of shifts	Adopt	Local linearity; elasticity ranges. Signs from LW-3 and LW-4	Any policy that shifts demand or supply (subsidies to buyers, entry barriers, mandates)	`statics.equilibrium_shift`
Shortage under a ceiling	Adopt	As above	PC-1, PC-2	`statics.ceiling_shortage`
Surplus under a floor (minimum wage)	Adopt	As above	PC-9, PC-10, PC-11	`statics.floor_surplus`
Quantity supplied under a held-down price	Adopt	As above, with a horizon-specific supply elasticity	PC-5	`statics.ceiling_quantity_supplied`
Tax incidence	Adopt	As above	TX-1, TX-2, TX-3; who bears a tax	`statics.tax_incidence`
Subsidy incidence	Adopt	As above	SB-1, SB-2, SB-8	`statics.subsidy_incidence`
Quantity traded under a wedge (exchanges forgone)	Adopt, counted not valued	As above	CR-5a, TX-2, OC-2 (compliance cost as a wedge)	`statics.quantity_under_wedge`

Method	Verdict	Assumes	Relevance	Module
Tariff: domestic price, output, use, imports	Adopt; pass-through as a range	As above; pass-through in [0, 1]	TX-7, TX-8, SB-10	`statics.tariff`
Quota rent and license value	Adopt (M0 once the gap is known)	The price gap as a range	EN-1, EN-6, EN-12, EN-13 (value of a right)	`statics.quota_rent` + `capital.present_value`
Spending on a good after a price change, (1+ε)·%ΔP	Adopt	As above	CR-4's means exception (whether spending on the burdened good rises or falls); TX-6	`statics.spending_change`
Cost pass-through up and down a chain	Bound: [0, cost share × input change]	Between no and full pass-through	RS-14, PC-12, TX-7 (steel users)	`statics.cost_pass_through`
Tax revenue with base response (the arithmetic behind the Laffer curve)	Adopt, first order; no claim about a revenue peak	Elasticity of the base to the net-of-tax rate, as a range	OC-1, TX-2, TX-9 (does revenue rise by less than the rate?)	`statics.revenue_change`
Cross-price effects (spillover to substitutes)	Adopt via `equilibrium_shift`	A demand shift in the substitute market = cross elasticity × price change	PC-8, RS-1, RS-4, RS-16	`equilibrium_shift` with a = cross elasticity × %ΔP
Black-market price bounds	Bound	Under a ceiling, the illegal premium lies between sellers' expected penalty per unit and what the marginal buyer will pay for the shrunken quantity, less the buyer's penalty (first order). An empty range means illegal trade does not pay, so imperfect enforcement fails	PC-6, RS-3	`bounds.black_market_premium`
Hotelling rule (an exhaustible resource's price rises at the rate of interest)	Bound only	No-arbitrage over time; owners expect to keep the resource and choose when to extract	Resource taxes and quotas (EN-, TX-12). Returns net-of-tax price growth minus r: negative brings extraction forward, positive defers it. Gives the direction and size of the timing pressure, not an extraction path	`bounds.hotelling_gap`
Rent-seeking dissipation (Tullock)	Bound: total lobbying spending lies between zero and the present value of the privilege to all contenders together	That contenders do not overpay in total	RS-7 (lobbying)	`bounds.rent_seeking_ceiling`

3. Valuation in money of gains and losses

Method	Verdict	Why
Consumer and producer surplus; deadweight loss (Harberger triangle) in money	**Exclude**	It treats the area under a demand curve as a sum of value. Value is ordinal and cannot be added across persons or measured in money (LW-1, HA ch. II and XIV). The annex counts the exchanges forgone (`quantity_under_wedge`) and stops there.
Cost-benefit analysis with net present value of welfare	**Exclude**	Same reason, plus interpersonal comparison. Present value of *money flows* (taxes, rents, asset prices) is adopted; present value of *welfare* is not.
Compensating and equivalent variation	**Exclude**	Same: a money measure of utility.
Social welfare functions, optimal tax (Ramsey, Mirrlees)	**Exclude**	Normative, and they need a welfare function. Out of scope (the handbook's analysis is positive).

4. Optimization, market structure and strategic models

Method	Verdict	Why
Utility maximization with calculus (Lagrangians, indifference curves)	**Exclude**	Cardinal or differentiable utility. CR-4 already gives every direction these models are used for, from ordinal ranking alone.
Cost minimization and production functions (Cobb-Douglas, CES)	**Exclude** (functional forms)	Cost shares are adopted as *inputs* (`cost_pass_through`); the forms are not.
Perfect competition as a benchmark	**Exclude**	An imaginary state (Mises's evenly rotating economy) that never obtains. The annex works with changes, not with distance from an ideal.
Monopoly pricing, Lerner index, markups $p = MC \cdot \varepsilon/(\varepsilon+1)$	**Exclude**	Needs a known demand curve and profit maximization at a point. Where a policy creates a single seller, the handbook's EN patterns give the direction.
Oligopoly models (Cournot, Bertrand, Stackelberg)	**Exclude**	Results depend on the model chosen, and each choice is an assumption about how rivals think.
Game theory: Nash equilibrium, mechanism design	**Exclude as computation; already praxeological where it is logic**	Dominant-strategy arguments (a prisoner's dilemma among sellers) are CR-4 applied to each actor. Equilibria that need common knowledge of payoffs are assumptions.
Search and matching models	**Exclude**	Functional forms for matching. Little's law (adopted) gives the waiting arithmetic without them.
Cobweb and hog-cycle models	**Exclude**	They assume a specific way of forming expectations. CR-10 handles timing.

5. Macroeconomic and system models

Method	Verdict	Why
Keynesian multipliers (spending multiplier, balanced-budget multiplier)	**Exclude**	They assume idle resources respond only to aggregate spending, and treat spending, not saving, as the source of income. That contradicts CR-8 (a tax-funded outlay is taken from other uses) and has no praxeological basis.
IS-LM, AD-AS, Phillips curve, Taylor rule	**Exclude**	Aggregate behavioral relations, estimated or assumed. They carry many assumptions and are contested.
DSGE and computable general equilibrium (CGE) models	**Exclude**	Hundreds of functional forms and calibrated parameters; they assume an equilibrium. They are the opposite of "few assumptions".
Input-output (Leontief) tables	**Exclude as a model; Input as data**	Fixed coefficients are an assumption. An input-output table's cost shares are adopted as *inputs* to `cost_pass_through`.
Growth accounting (Solow residual)	**Exclude**	A residual defined by a production function.

Method	Verdict	Why
Ricardian equivalence	**Exclude**	A behavioral claim (people offset deficits by saving), empirical and disputed. The debt arithmetic is adopted (`debt_path`); the behavioral claim is not.
Purchasing power parity (absolute or relative)	**Input**	A long-run tendency, not an identity. Usable as a range for the exchange-rate response in TX-8, never as a law.
Covered interest parity	**Bound**	An arbitrage condition that holds closely in practice. Usable as a bound on forward exchange rates, within a band for costs and capital controls; rarely needed by the handbook. Built as `bounds.covered_interest_parity`; there is no annex handler, because the ontology has no forward-rate variable.

6. Statistical and empirical methods

Method	Verdict	Why
Econometrics, regression, difference-in-differences, synthetic control	**Input**	They estimate elasticities and effects in past cases. They are the right source for the ranges the annex takes, and for outcome cards. They are not a way of deriving consequences. A cited estimate's confidence interval is a natural range.
Calibration and simulation	**Exclude**	Points chosen to fit, not ranges sourced.
Probability and expected value	**Adopt where it is arithmetic**	Expected penalties (above); the system's assumption probabilities (already in the method).
Forecasting from time series	**Input**	Counterfactual paths (the market rent without the policy) are inputs: a range, cited.

Summary

- **Adopted, in code (25 functions):**
  - M0, 14 functions: the equation of exchange and the velocity it implies; the Fisher relation; the inflation tax; debt dynamics; sectoral balances; present value, capitalized loss and price-path arithmetic; stock-flow arithmetic; doubling time; Little's law; expected penalty; quota rent.
  - M1, 11 functions: the supply and demand intersection; ceiling shortage, floor surplus and quantity supplied under a held-down price; tax and subsidy incidence; exchanges forgone; the tariff; spending change; cost pass-through (as a bound); tax revenue with base response.
- **Bounds only, in code (4 functions):** black-market premium, Hotelling timing, rent-seeking ceiling, covered interest parity.
- **Inputs, not methods:** econometric estimates, index numbers, input-output cost shares, purchasing power parity, forecasts.
- **Excluded:**
  - every money measure of welfare (surplus, deadweight loss in money, cost-benefit in welfare terms);
  - utility calculus and functional forms;
  - market-structure, oligopoly and equilibrium-game models;
  - Keynesian multipliers;
  - macro and general-equilibrium systems, growth accounting, Ricardian equivalence.

Appendix. Worked annex examples

Generated by the scripts in `math-augmented-praxeo/examples/`. All ranges are illustrative unless marked as from the case.

WE-001 math annex (illustrative ranges)

Generated by `examples/we001_annex.py`. Every elasticity, rate and duration is an illustrative range, not an estimate. The annex sizes three consequences that WE-001 signed and checks that each range agrees with the sign. It changes no sign.

Consequence	Praxeological sign	Annex range	Rung	Sign check	Basis
C-2 excess_demand (year 3)	increase	[0.008486, 0.1286]	M1	agrees	gap in year 3 = [0.08486, 0.1837] of the market rent
C-9 quantity_supplied (year 10)	decrease	[-0.4161, -0.04358]	M1	agrees	gap in year 10 = [0.1453, 0.4161] of the market rent
C-20 asset_value (at announcement)	decrease	[-6.775, -0.3121]	M0	agrees	change in building value, in years of today's rent

Assumptions, by consequence:

- C-2 excess_demand (year 3): ceiling [0.08486, 0.1837] below the clearing price; demand elasticity [-0.5, -0.1]; supply elasticity [0, 0.2]; elasticities roughly constant over the gap (first order). **First order is strained: the gap reaches 18%, beyond 15%; treat the range as indicative only.**
- C-9 quantity_supplied (year 10): price held [0.1453, 0.4161] below clearing; supply elasticity [0.3, 1] at the relevant horizon. **First order is strained: the gap reaches 42%, beyond 15%; treat the range as indicative only.**
- C-20 asset_value (at announcement): discount rate [0.04, 0.08]; expected duration [5, 15] years; gaps as given. **Borne by owners when the policy became expected (CR-7c); later buyers pay the lower price.**

Gap path (market minus capped rent, in units of today's rent): y1 [0.03, 0.07], y3 [0.09273, 0.225], y5 [0.1189, 0.3622], y10 [0.1952, 0.8185], y15 [0.2897, 1.491]

WE-013 math annex (illustrative ranges)

Generated by `examples/we013_annex.py`. Every elasticity, share and pass-through range is illustrative, not an estimate. The annex sizes six consequences that WE-013 signed and checks each range against the sign. It changes no sign.

Consequence	Praxeological sign	Annex range	Rung	Sign check	Basis
C-6 steel users, price_paid (domestic steel)	increase	[0.15, 0.25]	M1	agrees	tariff passed into the domestic price
C-2 domestic steelmakers, quantity_supplied	increase	[0.03, 0.25]	M1	agrees	supply response to the higher price
C-7 importers, quantity_traded (imports)	decrease	[-0.8026, -0.1129]	M1	agrees	use minus domestic output
C-27 steel users, quantity_traded (steel bought)	decrease	[-0.15, -0.03]	M1	agrees	demand response
C-25 tariff revenue vs static projection	decrease	[-0.8026, -0.1129]	M1	agrees	share of the static projection lost
C-24 construction buyers, price_paid	increase	[0, 0.025]	M1	indeterminate	steel's share of construction costs

Assumptions, by consequence:

- C-6 steel users, price_paid (domestic steel): tariff [0.25, 0.25]; import share [0.38, 0.42]; demand elasticity [-0.6, -0.2]; domestic supply elasticity [0.2, 1]; pass-through to the domestic price [0.6, 1]; first order
- C-2 domestic steelmakers, quantity_supplied: tariff [0.25, 0.25]; import share [0.38, 0.42]; demand elasticity [-0.6, -0.2]; domestic supply elasticity [0.2, 1]; pass-through to the domestic price [0.6, 1]; first order
- C-7 importers, quantity_traded (imports): tariff [0.25, 0.25]; import share [0.38, 0.42]; demand elasticity [-0.6, -0.2]; domestic supply elasticity [0.2, 1]; pass-through to the domestic price [0.6, 1]; first order. **First order is strained: the gap reaches 25%, beyond 15%; treat the range as indicative only.**
- C-27 steel users, quantity_traded (steel bought): tariff [0.25, 0.25]; import share [0.38, 0.42]; demand elasticity [-0.6, -0.2]; domestic supply elasticity [0.2, 1]; pass-through to the domestic price [0.6, 1]; first order
- C-25 tariff revenue vs static projection: tariff [0.25, 0.25]; import share [0.38, 0.42]; demand elasticity [-0.6, -0.2]; domestic supply elasticity [0.2, 1]; pass-through to the domestic price [0.6, 1]; first order. **Revenue falls short of the static projection (tariff x old imports) by the fall in imports.**
- C-24 construction buyers, price_paid: input price change [0.15, 0.25]; cost share [0.03, 0.1]; first order; between no and full pass-through

Annex pass: San Francisco Proposition I (1994)

Sizes attached to reported consequences by `mathpraxeo.bridge.annex_pass`. No consequence was changed.

Consequence	Actor class	Variable	Sign	Handler	Math rules	Annex range	Rung	Sign check
C-1	landlords.controlled	quantity_supplied	decrease	ceiling_quantity_supplied	MR-3, MR-4	[-0.341, -0.0129]	M1	agrees
C-3	tenants.controlled	price_paid	decrease	insiders_price	MR-3	[-0.341, -0.0429]	M0	agrees
C-8	tenants.prospective	price_substitute	increase	substitute_price	MR-5	[0.00455, 0.1]	M1	agrees

Sources of the ranges:

- market_growth: illustrative; to be sourced from pre-1994 San Francisco rent series
- cap_growth: 60% of regional CPI (case text) times a CPI range of 2% to 3.5% (illustrative)
- gap_year: horizon chosen for the annex (medium run)
- supply_elasticity: illustrative long-run elasticity of rental supply from the covered stock
- diverted_demand: illustrative: demand moved to uncovered units, as a share of their market
- substitute_demand_elasticity: illustrative
- substitute_supply_elasticity: illustrative

Notes:

- C-1: First order is strained: the gap reaches 34%, beyond 15%; treat the range as indicative only.
- C-3: Sitting tenants' price against the market path, as a share of the market price.

Premise step: San Francisco Proposition I (1994)

Annex evidence for dominance premises, put to a decider. The decider's answer is the premise's new probability; a proposal is never applied on its own. The frozen original prediction is unchanged.

Premise	Current p	Annex rung	Proposed p	Decider	Decider's p	Changed
`keepers_dominate(c1)`	0.60	M1	0.90	fixture:premise_answers.json	0.80	yes
`lockin_dominates(c1)`	0.65	M1	0.65	(not asked: no change proposed)	0.65	no

Evidence:

- `keepers_dominate(c1)`: annex M1: favoured size [150, 720] exceeds opposed size [18, 144] everywhere
- `lockin_dominates(c1)`: annex M1: ranges overlap ([150, 720] against [32, 180]); no change proposed

Consequences that differ in the annex-informed variant:

- none